Notes: Tate & Yang (JFE, 2015)

Female Leadership and Gender Equity: Evidence from Plant Closure

Contents

1 Big picture

- **Question.** Do women in leadership positions change workplace outcomes for other women, and can female leadership reduce gender wage gaps inside firms?
- **Core setting.** The paper uses worker-firm-plant matched administrative data from the U.S. Census Bureau and focuses on workers displaced by plant closures. Plant closures generate forced job changes that reduce concerns about voluntary sorting by gender.
- **Key outcome.** The main outcome is the change in a worker's log annual real wage from before plant closure to four quarters after reemployment.
- **Main comparison.** The strongest specifications compare men and women displaced from the same closing plant who move to the same new state-employer identification number (SEIN). This pair fixed effect holds fixed the closing shock and the new employer unit.
- **Main result.** Women experience roughly 4–5 percent larger wage losses than comparable men after displacement. The relative loss is substantially smaller when the hiring firm or hiring unit has more women in top leadership roles.
- **Critical mass.** The mitigation effect is strongest when women hold a majority of the top five leadership positions, suggesting that one woman in a senior role may not be enough to change culture.
- **Culture versus local interaction.** Firm-level female CEOs appear more important than female divisional leaders, and the effect does not concentrate in the CEO's home division. This points toward firm-wide culture rather than direct bargaining or local information.
- **Mechanism.** The effect is strongest for women displaced from male-led plants and from less competitive industries. This pattern is more consistent with female leaders reducing discrimination or demand-side bias than simply raising female productivity through amenities.
- **Economic takeaway.** Female leadership creates a positive externality for other women. Leadership diversity can affect firm culture, wage setting, and labor market outcomes below the top of the organization.

2 Incremental contribution of this paper

- **From representation to causal workplace outcomes.** Prior work documents that women are underrepresented in leadership and that gender wage gaps exist. Tate and Yang ask whether female leadership changes wage outcomes for other women.
- **Plant-closure design.** The paper advances the managerial-style and gender-gap literatures by using involuntary displacement from plant closures, rather than ordinary job changes, to reduce endogenous worker sorting.
- **Pair fixed effect comparison.** The paper compares male and female workers from the same closing plant who join the same hiring SEIN. This is a particularly sharp design because it holds fixed the origin shock and the destination employer.
- **Administrative data scale.** The study uses Census LEHD/LBD data to follow hundreds of thousands of workers across firms and plants, allowing a much richer analysis than survey-based leadership studies.
- **Leadership measured inside firms.** Instead of relying only on board seats or CEO names, the paper identifies the gender composition of leadership teams using the top five earners within a SEIN or firm.
- **Culture mechanism.** The paper distinguishes local division leadership from firm-level leadership. The fact that firm-level female CEOs matter more supports a culture interpretation.
- **Critical mass result.** The paper contributes evidence that majority female leadership has a stronger effect than small female representation, a key idea in diversity and governance debates.

- **Discrimination mechanism.** The paper connects female leadership to the mitigation of demand-side discrimination by showing stronger effects in concentrated labor markets, where firms have more wage-setting discretion.
- **Broader managerial-style channel.** The paper expands the corporate finance managerial-style literature beyond investment, leverage, and payout policy toward firm culture and labor-market treatment.

3 Data and measurement

3.1 Administrative data sources

- Plant-level information comes from the Longitudinal Business Database (LBD), which covers U.S. nonfarm establishments and identifies plant closures.
- Worker-level wages, employment, gender, race, age, and firm identifiers come from the Longitudinal Employer-Household Dynamics (LEHD) program.
- The sample uses 23 states available in the LEHD during the relevant period.
- Plant closures are observed between 1993 and 2001 so that workers can be observed before closure and followed after displacement.

Interpretation: The data allow the paper to observe workers, their old plants, their new employers, and the gender composition of leadership in both old and new firms.

3.2 Plant closures and displaced workers

- The core sample contains workers employed at a closing plant two quarters before the plant disappears.
- The pre-closure wage is measured before the closure, typically using annualized real wages over a pre-closure window.
- The post-closure wage is the annual real wage four quarters after the plant closure, conditional on reemployment.
- The dependent variable in most displaced-worker tests is the log change in annual wages around the closure.

Interpretation: Plant closure creates an involuntary shock. The design reduces selection concerns that arise when men and women voluntarily move to different firms for different reasons.

3.3 Leadership measures

- **Top five earners:** leadership in a SEIN or firm is measured using the top five highest-paid workers.
- **Percentage female leaders:** the percentage of women among the top five earners.
- **Majority female leadership:** an indicator that more than 50 percent of the top five leaders are women.
- **Female CEO:** an indicator that the top earner in the firm is female.
- **Female divisional CEO:** an indicator that the top earner in the hiring SEIN is female.

Interpretation: The leadership measures are wage-rank proxies, not job-title measures. They are useful in administrative data but may contain some noise.

3.4 Worker and firm controls

- Worker controls include race indicators, log age, log pre-closure wage, manager status, and log tenure.
- Firm and plant controls include plant fixed effects, new SEIN fixed effects, and closing plant-new SEIN pair fixed effects.
- Some robustness checks interact pair fixed effects with wage-bin fixed effects to compare workers closer together in the pre-closure wage distribution.

Interpretation: Controls and fixed effects are designed to remove differences in baseline ability, origin plant shocks, destination firm shocks, and post-closure job choices.

4 Models and empirical specifications

4.1 Wage-level model in random worker sample

The paper first documents a cross-sectional relation between female leadership and the gender wage gap:

$$\log(Wage_{i,j,t}) = \alpha + \beta Female_i + \gamma FemaleLeadership_{j,t} + \delta(Female_i \times FemaleLeadership_{j,t}) + X_i'\theta + FE + \varepsilon_{i,j,t}.$$

Interpretation: A positive interaction δ means female leadership is associated with a smaller gender wage gap in wage levels.

4.2 Displaced-worker wage-change model

The main model uses wage changes around plant closure:

$$\Delta \log(Wage_i) = \alpha + \beta Female_i + \gamma FemaleLeadership_{h(i)} + \delta(Female_i \times FemaleLeadership_{h(i)}) + X_i'\theta + FE + \varepsilon_i.$$

- $\Delta \log(Wage_i)$ is the difference between post-closure and pre-closure log annual wage.
- $h(i)$ is the hiring SEIN or hiring firm.
- The key coefficient is δ.
- A positive δ means female leadership reduces the relative wage loss of displaced women.

Interpretation: The wage-change model removes time-invariant worker differences. The pair fixed effect compares men and women experiencing the same closure shock and entering the same employer unit.

4.3 Pair fixed effect specification

$$\Delta \log(Wage_i) = \alpha_{ClosingPlant \times HiringSEIN} + \beta Female_i + \delta(Female_i \times FemaleLeadership_h) + X_i'\theta + \varepsilon_i.$$

Interpretation: This is the key identification specification. It holds fixed the old plant and new employer unit, so the gender coefficient and leadership interaction are identified within the same job-change group.

4.4 CEO versus divisional leadership

$$\Delta \log(Wage_i) = \alpha_{pair} + \beta Female_i + \delta_1(Female_i \times FemaleCEO_f) + \delta_2(Female_i \times FemaleDivisionalCEO_s) + X_i'\theta + \varepsilon_i.$$

Interpretation: If the effect operates through direct bargaining or local information, divisional leadership should matter. If it operates through firm-wide culture, firm-level CEOs should matter more.

4.5 Sorting and quantity effect model

$$1\{FemaleCEO_{h(i)} = 1\} = \alpha + \beta Female_i + X_i'\theta + FE + \varepsilon_i.$$

Interpretation: This tests whether displaced women sort in larger quantities into female-led firms. A positive coefficient indicates that women are more likely to move to female-led employers.

4.6 Industry-competition heterogeneity

$$\Delta \log(Wage_i) = \alpha_{pair} + \beta_q Female_i + \delta_q(Female_i \times FemaleCEO_h) + X_i'\theta + \varepsilon_i,$$

where q indexes quartiles of industry concentration.

Interpretation: If female leaders mitigate discrimination, the effect should be stronger in concentrated industries where firms have more discretion to pay noncompetitive wages.

5 Identification and empirical design

5.1 Core identification problem

Women may choose different jobs than men, and female managers may be more likely to work in already female-friendly firms. Cross-sectional wage gaps can therefore conflate leadership effects, worker preferences, family responsibilities, unobserved ability, job amenities, and firm culture.

5.2 Plant closures as involuntary displacement

The plant-closure design creates a sample of workers forced to search for new jobs. This reduces concerns that observed wage changes are driven by voluntary timing of job changes or endogenous departures.

5.3 Same-origin, same-destination comparison

The strongest design compares men and women from the same closing plant who move to the same hiring SEIN. This removes:

- closure-specific shocks,
- local labor-market shocks faced by the closing plant,
- destination firm-unit wage policies common to both genders,
- broad differences in post-closure job choices.

5.4 Pre-closure wage and wage-bin robustness

Pre-closure wages summarize worker quality, occupation, firm-specific human capital, and prior wage setting. Wage-bin by pair fixed effects further ensure that identification is not driven by comparisons between men and women far apart in the wage distribution.

5.5 Culture versus direct interaction

The paper separates firm-level leadership from divisional leadership. The finding that firm-level female CEOs matter more than divisional female leaders suggests that leadership changes organizational culture or policies rather than simply affecting direct negotiations.

5.6 Mechanism tests

The paper uses two main mechanism tests:

- **Closing plant leadership:** women coming from female-led plants have less to gain from female-led hiring firms, consistent with prior exposure to female-friendly cultures.
- **Industry concentration:** the female CEO effect is strongest for workers displaced from concentrated industries, consistent with discrimination or wage-setting discretion.

6 Section-by-section study guide

- **Introduction.** Sets up female leadership as a potential mechanism for changing firm culture and reducing gender inequity.
- **Data.** Describes the LBD and LEHD data, plant closures, worker-firm matches, wage measurement, and leadership proxies.
- **Random sample wage-level analysis.** Shows that female leadership is associated with smaller gender wage gaps in wage levels.
- **Plant-closure strategy.** Uses involuntary displacement and pair fixed effects to compare men and women facing the same closure and new employer.
- **Female leadership in hiring firms.** Shows that female leadership reduces displaced women's relative wage losses.
- **CEO versus division leadership.** Distinguishes firm-wide culture from local direct manager effects.
- **Sorting and mechanisms.** Tests whether women sort into female-led firms and whether results are strongest where discrimination is more plausible.
- **Conclusion.** Interprets female leadership as a positive externality and a channel through which managerial style affects firm culture.

7 Tables and figures

7.1 Tables 1 and 2: Plant and worker summary statistics

Read:

- Table 1 describes random plants, closing plants, and the LBD-LEHD matched closing-plant sample.
- The matched closing-plant sample is smaller and underrepresents multi-unit firms because workers must be matchable to unique establishments.
- Table 2 describes random worker-quarters from LEHD.
- Women earn substantially less than men in the random worker sample and are more likely to work in firms with female leadership.

Detailed interpretation: Table 1 tells readers how selected the plant-closure sample is. The matched sample is not a random cross-section of all plants. Table 2 establishes the baseline gender wage gap and leadership distribution in the broader worker sample.

Economic message: The paper's empirical strategy starts from large baseline gender wage differences but relies on plant closures and fixed effects to reduce selection concerns.

Table 1
Summary statistics: plant level.
The table reports summary statistics of a random sample of closing plants from the US Bureau of the Census's Longitudinal Business Database (LBD), a random sample of non-closing plants from the LBD, and the subsample of closing plants from the LBD that we match with worker-level data from the Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) program. The table also reports the corresponding statistics for the subsamples of plants from multi-unit firms. We define multi-unit firms as firms that operate at least two distinct plants. Standard deviations are reported in parentheses for continuous variables. * Denotes subsamples on which we cannot report distributional statistics due to disclosure risk (some partitions contain too few firms). SIC=standard industrial classification.

	All firms			Multi-unit firms only		
	Random plants in the LBD (N=655,929)	Closing plants in the LBD (N=143,370)	Closing plants in the LBD matched with the LEHD (N=12,439)	Random plants in the LBD (N=383,238)	Closing plants in the LBD (N=70,811)	Closing plants in the LBD matched with the LEHD (N=1,850)
Plant employees	194	188	134	202	187	142
	(514)	(647)	(292)	(473)	(565)	(224)
Firm employees	25,765	22,084	4,780	43,968	44,521	31,379
	(83,464)	(57,124)	(26,992)	(105,480)	(74,912)	(63,789)
Payroll (in thousands of dollars)	$6,830	$5,299	$2,333	$7,590	$6,676	$3,703
	($383,230)	($66,606)	($6,709)	($178,102)	($92,809)	($9,611)
Percent of multi-unit firms	0.58	0.49	0.15	–	–	–
Percent of diversified firms	0.42	0.39	0.10	0.71	0.79	0.69
Industry distribution						
SIC=1	0.05	0.04	0.09	0.02	0.02	
SIC=2	0.08	0.08	0.08	0.09	0.08	
SIC=3	0.10	0.08	0.07	0.10	0.09	
SIC=4	0.06	0.07	0.05	0.08	0.08	*
SIC=5	0.29	0.27	0.28	0.36	0.30	
SIC=6	0.06	0.09	0.04	0.07	0.10	
SIC=7	0.13	0.19	0.24	0.13	0.18	
SIC=8	0.21	0.16	0.13	0.15	0.13	
Geographic distribution						
LEHD state	0.55	0.57	–	0.55	0.57	–
Northeast	0.22	0.22	0.08	0.21	0.22	0.09
Midwest	0.25	0.21	0.16	0.25	0.22	0.18
South	0.23	0.24	0.23	0.24	0.24	0.26
Southwest	0.12	0.13	0.19	0.12	0.12	0.19
West	0.14	0.16	0.29	0.14	0.15	0.22
Rocky Mountains	0.04	0.03	0.05	0.04	0.03	0.06
Yearly distribution						
1994	0.10	0.08	0.08	0.10	0.07	0.05
1995	0.11	0.08	0.10	0.10	0.08	0.07
1996	0.11	0.11	0.12	0.11	0.11	0.13
1997	0.11	0.10	0.09	0.11	0.10	0.07
1998	0.11	0.11	0.13	0.12	0.11	0.12
1999	0.12	0.12	0.12	0.12	0.14	0.10
2000	0.12	0.12	0.14	0.12	0.13	0.22
2001	0.12	0.21	0.14	0.12	0.17	0.17

(a) Table 1: Summary statistics: plant level.

selection of closing plants into our sample. our sample, we regress the natural logarithm of annual

Table 2
Summary statistics: random sample of workers.
This table reports summary statistics for a random sample of workers from the US Bureau of the Census's Longitudinal Employer-Household Dynamics (LEHD) data. Annual wage is the mean real wage over the preceding four quarters multiplied by four. To be included in the annual wage computation, a quarter cannot be the worker's first or last quarter in his or her current employment spell. Tenure is artificially set to zero for the first year each state appears in the LEHD universe. Education is imputed using an algorithm constructed by the LEHD program. *% x Leaders* measures the percentage of workers of type x among the top five earners in the worker's state employer identification number (SEIN).

	Overall			Men			Women		
	N	Mean	Standard deviation	N	Mean	Standard deviation	N	Mean	Standard deviation
Worker characteristics									
Annual wage	235,822	35,145	92,402	127,405	41,683	102,060	108,417	27,461	76,086
Age	235,822	41.34	11.10	127,405	41.37	11.14	108,417	41.30	11.07
Tenure (years)	235,822	3.51	2.61	127,405	3.49	2.60	108,417	3.70	2.60
Education (years)	235,822	13.78	2.60	127,405	13.63	2.74	108,417	13.97	2.41
Female	235,822	0.46							
Race									
White	235,822	0.72		127,405	0.73		108,417	0.72	
Black	235,822	0.10		127,405	0.09		108,417	0.11	
Asian	235,822	0.04		127,405	0.04		108,417	0.04	
Hispanic	235,822	0.09		127,405	0.09		108,417	0.08	
Other	235,822	0.05		127,405	0.06		108,417	0.04	
Foreign	235,822	0.14		127,405	0.16		108,417	0.13	
Native to state	235,822	0.44		127,405	0.43		108,417	0.46	
Firm-unit characteristics									
% Female Leaders	235,822	0.15	0.21	127,405	0.11	0.17	108,417	0.20	0.24
% Black Leaders	235,822	0.03	0.10	127,405	0.03	0.09	108,417	0.04	0.11
% Hispanic Leaders	235,822	0.01	0.06	127,405	0.01	0.07	108,417	0.01	0.06

(b) Table 2: Summary statistics: random sample of workers.

Tables 1 and 2: Data construction and baseline worker/plant characteristics.

7.2 Tables 3 and 4: Female leadership in wage levels and displaced-worker characteristics

Read:

- Table 3 shows that women earn less than men in wage levels, but the gap is smaller in SEINs with more women among top earners.

- The interaction between female worker and female leadership is positive in the main cross-sectional wage regressions.
- Table 4 describes the displaced-worker sample and the closing plant-new SEIN groups used for identification.
- Men and women in the displaced-worker matched groups look similar on many observables after conditioning on the job-change group.

Detailed interpretation: Table 3 establishes the cross-sectional fact: female leadership is associated with narrower gender wage gaps. Table 4 provides the bridge to the causal design by describing the involuntary displacement sample.

Economic message: Female leadership is correlated with wage equality, but the key question is whether that correlation survives a sharper displaced-worker design.

Table 3
Female leadership and wage levels.
The dependent variable is the natural logarithm of the annual wage (defined in Table 2). The sample in Column 1 is a random sample of worker-quarters from US firms in 23 states covered by the Longitudinal Employer-Household Dynamics (LEHD) Program of the US Bureau of the Census. Columns 2–4 exclude workers who are among the top five earners in their state employer identification number (SEIN). The omitted race category is white. *Age* is worker age. *Tenure* is measured as the number of quarters that a worker has spent in the firm. *Foreign* is an indicator for workers born outside the United States. *Native* is an indicator for workers who were born in the state in which they are currently employed. We define diversified firms (*Diversified*) as firms that operate in at least two distinct two-digit standard industrial classification codes. *Firm Employment* is measured as the total number of workers for the entire firm (across all its plants). *Female Top Leader* is an indicator equal to one if the top earner in the worker's SEIN is female. *% Female Leaders* measures the percentage of women among the top five earners in the worker's SEIN. Standard errors are clustered by SEIN and are reported in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

Independent variable	(1)	(2)	(3)	(4)
Race = black	−0.209***	−0.194***	−0.195***	−0.194***
	(0.004)	(0.004)	(0.004)	(0.004)
Race = Asian	−0.034***	−0.029***	−0.029***	−0.029***
	(0.009)	(0.008)	(0.008)	(0.008)
Race = Hispanic	−0.282***	−0.266***	−0.267***	−0.266***
	(0.005)	(0.005)	(0.005)	(0.005)
Race = other minorities	−0.036***	−0.033***	−0.033***	−0.033***
	(0.006)	(0.006)	(0.006)	(0.006)
Foreign	−0.102***	−0.094***	−0.094***	−0.094***
	(0.006)	(0.005)	(0.005)	(0.005)
Native	−0.123***	−0.115***	−0.115***	−0.115***
	(0.003)	(0.003)	(0.003)	(0.003)
Education	0.037***	0.035***	0.035***	0.035***
	(0.001)	(0.001)	(0.001)	(0.001)
ln(*Age*)	0.296***	0.261***	0.262***	0.262***
	(0.006)	(0.006)	(0.006)	(0.006)
ln(*Tenure*)	0.099***	0.096***	0.097***	0.096***
	(0.002)	(0.002)	(0.002)	(0.002)
Diversified	0.016***	0.013**	0.014**	0.013**
	(0.006)	(0.006)	(0.006)	(0.006)
ln(*Firm Employment*)	0.029***	0.036***	0.037***	0.036***
	(0.001)	(0.001)	(0.001)	(0.001)
Female	−0.297***	−0.303***	−0.280***	−0.308***
	(0.004)	(0.004)	(0.004)	(0.004)
% Female Leaders		−0.276***		
		(0.012)		
(*% Female Leaders*) × (*Female*)		0.226***		
		(0.012)		
Female Top Leader			−0.074***	0.014**
			(0.007)	(0.007)
(*Female Top Leader*) × (*Female*)			0.056***	−0.018**
			(0.008)	(0.009)
% Female Leaders > 0				−0.047***
				(0.005)
% Female Leaders > 20				−0.058***
				(0.007)
% Female Leaders > 40				−0.068***
				(0.007)
% Female Leaders > 60				−0.07***
				(0.011)
(*% Female Leaders* > 0) × (*Female*)				0.056***
				(0.006)
(*% Female Leaders* > 20) × (*Female*)				0.071***
				(0.008)
(*% Female Leaders* > 40) × (*Female*)				0.032***
				(0.012)
(*% Female Leaders* > 60) × (*Female*)				0.021
				(0.017)
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.340	0.355	0.352	0.355
N	235,822	230,729	230,729	230,729

(a) Table 3: Female leadership and wage levels.

Table 4
Summary statistics: displaced workers.
This table reports summary statistics for a sample of workers from the US Bureau of the Census's Longitudinal Employer-Household Dynamics (LEHD) data matched to closing plants in the Census Bureau's Longitudinal Business Database (LBD). Annual wage is the mean real wage over the preceding four quarters multiplied by four. To be included in the annual wage computation, a quarter cannot be the worker's first or last quarter in his or her current employment spell. Tenure is artificially set to zero for the first year each state appears in the LEHD universe. Education is imputed using an algorithm constructed by the LEHD program. *% x Leaders* measures the percentage of workers of type *x* among the top five earners in the worker's state employer identification number (SEIN). Δ*State* (*Industry*) is an indicator equal to one if the worker's new job four quarters after plant closure is in a new state (two-digit standard industrial classification).

	Workers displaced from closing plants						Closing plant, new SEIN groups		
	Overall		Men		Women			Men	Women
	N	Mean	*N*	Mean	*N*	Mean	*N*	Mean	Mean
Worker characteristics									
Annual wage	461,449	29,933	272,757	34,303	188,692	23,615	15,830	35,463	23,855
Age	461,449	39.68	272,757	39.55	188,692	39.87	15,830	40.20	39.87
Tenure (years)	461,449	2.57	272,757	2.56	188,692	2.65	15,830	2.55	2.51
Education (years)	461,449	13.66	272,757	13.47	188,692	13.95	15,830	13.54	13.79
Female	461,449	0.41							
Race									
White	461,449	0.68	272,757	0.67	188,692	0.69	15,830	0.67	0.68
Black	461,449	0.10	272,757	0.09	188,692	0.12	15,830	0.09	0.09
Asian	461,449	0.04	272,757	0.04	188,692	0.05	15,830	0.04	0.05
Hispanic	461,449	0.12	272,757	0.14	188,692	0.10	15,830	0.12	0.11
Other	461,449	0.06	272,757	0.06	188,692	0.05	15,830	0.08	0.08
Foreign	461,449	0.14	272,757	0.21	188,692	0.16	15,830	0.21	0.17
Native to state	461,449	0.42	272,757	0.41	188,692	0.44	15,830	0.38	0.41
Δ*State*	461,449	0.02	272,757	0.03	188,692	0.02	n.a.	n.a.	n.a.
Δ*Industry*	461,449	0.33	272,757	0.34	188,692	0.31	n.a.	n.a.	n.a.
Firm-unit characteristics									
% Female Leaders	438,298	0.20	259,078	0.14	179,220	0.29	n.a.	n.a.	n.a.
% Black Leaders	438,298	0.03	259,078	0.03	179,220	0.04	n.a.	n.a.	n.a.
% Hispanic Leaders	438,298	0.04	259,078	0.05	179,220	0.04	n.a.	n.a.	n.a.

(b) Table 4: Summary statistics: displaced workers.

Tables 3 and 4: From wage-level correlations to the displacement sample.

7.3 Tables 5 and 6: Sorting diagnostics and main displaced-worker effect

Read:

- Table 5 compares men and women hired into firms with female versus male top earners.
- The most notable cross-group difference is annual wage; many other observable differences are modest.
- Table 6 is the main displaced-worker regression table.
- Women lose roughly 4–5 percent more than men after plant closure.

- Female leadership in the hiring SEIN significantly reduces the relative wage loss of women, especially when women are a majority of the top five leaders.

Detailed interpretation: Table 5 addresses whether women hired by female-led firms are simply observably different. Table 6 then uses fixed effects and controls to estimate the key treatment interaction. The main result is that majority female leadership cuts the displaced women's relative wage loss by roughly half.

Economic message: Female leadership does not merely correlate with better cross-sectional pay; it mitigates losses among women forced into new jobs after plant closure.

quality. However, we do not see differences n and women within job change groups in

to work against our main hypothesis. We re conclusions when we interact the observable

Table 5
Summary statistics: displaced worker traits by gender of the top earner in the hiring firm.

The table reports sample means for workers from the US Bureau of the Census's Longitudinal Employer-Household Dynamids (LEHD) data matched to closing plants in the Census Bureau's Longitudinal Business Database (LBD). The sample in the first (last) two columns is workers who find new jobs in firms with a female (male) top earner. Annual wage is the mean real wage over the preceding four quarters multiplied by four. To be included in the annual wage computation, a quarter cannot be the worker's first or last quarter in his or her current employment spell. Tenure is artificially set to zero for the first year each state appears in the LEHD universe. Education is imputed using an algorithm constructed by the LEHD program. ***, **, and * in Column 3 indicate significance of the difference in means between men in male- and female-led firms at the 1%, 5%, or 10% level, respectively. *** in Column 4 indicates significance of the difference in means between women in male- and female-led firms at the 1% level.

	Female top earner		Male top earner	
	Men (N=8,165) (1)	Women (N=12,506) (2)	Men (N=85,224) (3)	Women (N=54,747) (4)
Annual wage	28,638	21,949	32,794***	23,486***
Age	39.18	40.52	39.68	40.07
Tenure (years)	2.36	2.51	2.57	2.65
Education (years)	13.37	14.01	13.37*	14.03
Race				
White	0.62	0.70	0.66**	0.70
Black	0.10	0.12	0.09	0.11
Asian	0.04	0.05	0.03	0.04
Hispanic	0.17	0.09	0.15	0.09
Other	0.07	0.04	0.06	0.05
Foreign	0.24	0.15	0.21*	0.15
Native to state	0.40	0.46	0.43**	0.48

(a) Table 5: Worker traits by gender of the hiring firm's top earner.

Table 6
Female leadership and wage changes among displaced workers.

The dependent variable is the difference in the natural logarithms of the pre- and post-plant closure wage. The pre-closure wage is the annual wage (defined in Table 2) two quarters prior to plant closure, and the post-closure wage is the annual wage four quarters following the plant closure. The omitted race category is white. *Age* is worker age. *Wage* is the pre-closure annual wage. *Tenure* is measured as the number of quarters that a worker has spent in the firm. All worker-level variables are measured two quarters prior to plant closure. *Manager* is defined as the highest paid employee in the plant. *% Female Leaders* is the percentage of females among the top five earners in the worker's post-closure employer state employer identification number (SEIN). Standard errors are clustered by closing plant and are reported in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

	Displacement costs		Female leadership		
Independent variable	(1)	(2)	(3)	(4)	(5)
Race = black	−0.042***	−0.034***	−0.032***	−0.032***	−0.037***
	(0.003)	(0.003)	(0.004)	(0.004)	(0.005)
Race = Asian	−0.003	−0.014***	−0.016***	−0.016***	−0.015***
	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)
Race = Hispanic	−0.032***	−0.031***	−0.029***	−0.029***	−0.031***
	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Race = other minorities	−0.015***	−0.014***	−0.013***	−0.013***	−0.013***
	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)
ln(*Age*)	−0.071***	−0.072***	−0.069***	−0.069***	−0.068***
	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
ln(*Wage*)	−0.130***	−0.116***	−0.128***	−0.128***	−0.139***
	(0.004)	(0.004)	(0.006)	(0.006)	(0.006)
Manager	0.020***	0.026***	0.036***	0.036***	0.042***
	(0.007)	(0.008)	(0.010)	(0.010)	(0.011)
ln(*Tenure*)	−0.005***	−0.015***	−0.014***	−0.014***	−0.014***
	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Female	−0.052***	−0.037***	−0.041***	−0.040***	−0.045***
	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)
(*% Female Leaders*) × (*Female*)			0.015*		
			(0.009)		
% Female Leaders > 50					−0.021
					(0.015)
(*% Female Leaders* > 50) × (*Female*)				0.017***	0.021***
				(0.006)	(0.007)
Plant fixed effects	Yes	No	No	No	Yes
Plant-new SEIN pair fixed effects	No	Yes	Yes	Yes	No
New SEIN fixed effects	No	No	No	No	Yes
Adjusted R^2	0.149	0.476	0.532	0.532	0.456
N	359,537	359,537	256,881	256,881	256,881

(b) Table 6: Female leadership and wage changes among displaced workers.

Tables 5 and 6: Sorting diagnostics and main wage-change result.

7.4 Tables 7 and 8: Heterogeneity and CEO versus divisional leadership

Read:

- Table 7 shows the female-leadership effect by age and wage groups.
- The mitigation effect is strongest for women above age 35 and for women in the lower-middle wage distribution.
- Table 8 shows that female firm-level CEOs reduce the wage gap, while female divisional CEOs are not the main driver.
- The CEO result holds in single-division and multi-division firm samples.

Detailed interpretation: Table 7 helps distinguish mechanisms. If the effect were mainly about productivity-enhancing family-friendly amenities, one might expect stronger effects among younger women. Instead, effects are strong among older women and lower-middle wage groups. Table 8 points to broad culture because firm-level CEOs matter more than divisional leaders.

Economic message: Female leadership appears to change organizational culture and wage-setting norms, not just local bargaining with a direct manager.

Table 7
Female leadership and wage changes by age and wage groups.

The dependent variable is the difference in the natural logarithms of the pre- and post-plant closure wage. The pre-closure wage is the annual wage (defined in Table 2) two quarters prior to plant closure, and the post-closure wage is the annual wage four quarters following the plant closure. Worker and firm characteristics are the control variables from Table 5: race indicators (black, Asian, Hispanic, and other minorities), ln(*Wage*), *Manager*, and ln(*Tenure*). *% Female Leaders* is the percentage of females among the top five earners in the worker's post-closure employer state employer identification number (SEIN). All standard errors are clustered by closing plant and are reported in parentheses. *, **, and *** represent significance at 10%, 5%, and 1% level, respectively.

	Age breakouts		Wage breakouts	
Independent variable	(1)	(2)	(3)	(4)
(*Female*) × (*Age* < 25)	−0.065***	−0.057***		
	(0.006)	(0.007)		
(*Female*) × (25 ≤ *Age* < 35)	−0.056***	−0.060***		
	(0.004)	(0.004)		
(*Female*) × (35 ≤ *Age* < 45)	−0.033***	−0.038***		
	(0.003)	(0.003)		
(*Female*) × (45 ≤ *Age* < 55)	−0.033***	−0.041***		
	(0.003)	(0.004)		
(*Female*) × (*Age* ≥ 55)	−0.028***	−0.033***		
	(0.005)	(0.006)		
(*Female*) × (*Age* < 25) × (*% Female Leaders* > 50)	−0.008	−0.014		
	(0.014)	(0.016)		
(*Female*) × (25 ≤ *Age* < 35) × (*% Female Leaders* > 50)	0.013	0.015		
	(0.009)	(0.010)		
(*Female*) × (35 ≤ *Age* < 45) × (*% Female Leaders* > 50)	0.017**	0.024***		
	(0.007)	(0.008)		
(*Female*) × (45 ≤ *Age* < 55) × (*% Female Leaders* > 50)	0.025***	0.033***		
	(0.007)	(0.009)		
(*Female*) × (*Age* ≥ 55) × (*% Female Leaders* > 50)	0.025***	0.031***		
	(0.009)	(0.011)		
(*Female*) × (*Wage* < 20K)			−0.043***	−0.046***
			(0.003)	(0.004)
(*Female*) × (20K ≤ *Wage* < 40K)			−0.024***	−0.029***
			(0.004)	(0.004)
(*Female*) × (40K ≤ *Wage* < 60K)			−0.017***	−0.022***
			(0.005)	(0.006)
(*Female*) × (60K ≤ *Wage* < 100K)			−0.030***	−0.037***
			(0.009)	(0.011)
(*Female*) × (*Wage* ≥ 100K)			−0.012	−0.028
			(0.026)	(0.030)
(*Female*) × (*Wage* < 20K) × (*% Female Leaders* > 50)			0.007	0.008
			(0.008)	(0.009)
(*Female*) × (20K ≤ *Wage* < 40K) × (*% Female Leaders* > 50)			0.026***	0.034***

(a) Table 7: Female leadership and wage changes by age and wage groups.

Table 8
Female chief executive officer (CEO) and wage changes for displaced workers.

The dependent variable is the difference in the natural logarithms of the pre- and post-plant closure wage. The pre-closure wage is the annual wage (defined in Table 2) two quarters prior to plant closure, and the post-closure wage is the annual wage four quarters following the plant closure. The omitted race category is white. *Age* is worker age. *Wage* is the pre-closure annual wage. *Tenure* is measured as the number of quarters that a worker has spent in the firm. All worker-level variables are measured two quarters prior to plant closure. *Manager* is defined as the highest paid employee in the plant. *Female CEO* indicates that the top earner in the hiring firm (FIRMID) is female. *Female Divisional CEO* indicates that the top earner in the hiring state employer identification number (SEIN) is female. SD is the subsample of single-division (SEIN) firms. MD is the subsample of multi-division (SEIN) firms. All standard errors are clustered by closing plant and are reported in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

	Female CEO		Firm versus divisional CEO	
Independent variable	(1)	(2)	(3)	(4)
Race = black	−0.029***	−0.032***	−0.033***	−0.022***
	(0.006)	(0.007)	(0.007)	(0.008)
Race = Asian	−0.011*	−0.011	−0.003	−0.025**
	(0.007)	(0.008)	(0.008)	(0.013)
Race = Hispanic	−0.026***	−0.026***	−0.035***	−0.015
	(0.005)	(0.006)	(0.004)	(0.010)
Race = other minorities	−0.011***	−0.011***	−0.012***	−0.009*
	(0.003)	(0.004)	(0.005)	(0.005)
ln(*Age*)	−0.072***	−0.073***	−0.079***	−0.062***
	(0.005)	(0.007)	(0.007)	(0.009)
ln(*Wage*)	−0.122***	−0.127***	−0.127***	−0.123***
	(0.008)	(0.009)	(0.007)	(0.018)
Manager	0.023*	0.029**	0.029**	0.030
	(0.012)	(0.014)	(0.013)	(0.027)
ln(*Tenure*)	−0.014***	−0.014***	−0.014***	−0.013***
	(0.002)	(0.002)	(0.003)	(0.004)
Female	−0.040***	−0.042***	−0.040***	−0.042***
	(0.003)	(0.004)	(0.004)	(0.006)
Female CEO		−0.028		
		(0.022)		
(*Female CEO*) × (*Female*)	0.021***	0.022**	0.016*	0.040**
	(0.008)	(0.009)	(0.009)	(0.019)
(*Female Divisional CEO*) × (*Female*)				−0.009
				(0.015)
Sample	All	All	SD	MD
Plant fixed effects	No	Yes	No	No
Plant-new SEIN pair fixed effects	Yes	No	Yes	Yes
New SEIN fixed effects	No	Yes	No	No
Adjusted R^2	0.534	0.497	0.573	0.427
N	160,642	160,642	99,583	53,767

(b) Table 8: Female CEO and wage changes for displaced workers.

Tables 7 and 8: Heterogeneity and firm-level leadership.

7.5 Tables 9 and 10: Closing-plant leadership and quantity effects

Read:

- Table 9 splits displaced workers by leadership at the closing plant or closing firm.
- Women from male-led plants benefit most from moving to female-led hiring firms.
- Women from female-led origins already face smaller gender gaps and therefore gain less from female-led destinations.
- Table 10 tests whether female workers sort in greater numbers into female-led firms.
- Female workers are more likely to join female-led hiring firms, consistent with a higher demand for female workers by female-led firms.

Detailed interpretation: Table 9 shows that female-friendly culture at the origin firm already affects women. Table 10 shifts from wages to quantities and rejects a pure supply-side preference story: if women simply preferred female-led firms, the wage effect would tend to go the other way.

Economic message: Female leadership changes both prices and quantities in the labor market: it raises relative wages for women and increases the likelihood that women join female-led firms.

Table 9
Female leadership in closing plant and wage changes for displaced workers.

The dependent variable is the difference in the natural logarithms of the pre- and post-plant closure wage. The pre-closure wage is the annual wage (defined in Table 2) two quarters prior to plant closure and the post-closure wage is the annual wage four quarters following the plant closure. The sample excludes workers who are among the top five earners in their closing plants and includes only workers from closing and hiring firms for which we observe all state employer identification numbers (SEINs) in the US Bureau of the Census's Longitudinal Employer-Household Dynamics (LEHD) data. The omitted race category is white. *Age* is worker age. *Wage* is the pre-closure annual wage. *Tenure* is measured as the number of quarters that a worker has spent in the firm. All worker-level variables are measured two quarters prior to plant closure. *Manager* is defined as the highest paid employee in the plant. *Female CEO* indicates that the top earner in the hiring firm (FIRMID) is female. All standard errors are clustered by closing plant and are reported in parentheses. *, **, and *** represent significance at 10%, 5%, and 1% level, respectively.

	> 50% female top 5 closing plant	≤ 50% female top 5 closing plant	Female CEO closing firm	Male CEO closing firm
Independent variable	(1)	(2)	(3)	(4)
Race = black	−0.029***	−0.022***	−0.037***	−0.021***
	(0.010)	(0.006)	(0.011)	(0.005)
Race = Asian	0.017	−0.012	0.026	−0.014*
	(0.015)	(0.008)	(0.016)	(0.008)
Race = Hispanic	−0.018*	−0.022***	−0.011	−0.023***
	(0.011)	(0.006)	(0.010)	(0.006)
Race = other minorities	0.008	−0.011***	−0.001	−0.010**
	(0.010)	(0.004)	(0.010)	(0.004)
ln(*Age*)	−0.068***	−0.074***	−0.077***	−0.073***
	(0.010)	(0.007)	(0.011)	(0.007)
ln(*Wage*)	−0.091***	−0.133***	−0.097***	−0.131***
	(0.008)	(0.013)	(0.009)	(0.013)
ln(*Tenure*)	−0.020***	−0.014***	−0.014***	−0.015***
	(0.006)	(0.003)	(0.005)	(0.003)
Female	−0.018**	−0.040***	−0.019**	−0.039***
	(0.008)	(0.004)	(0.008)	(0.004)
(*Female CEO*) × (*Female*)	−0.011	0.028***	0.001	0.023**
	(0.015)	(0.010)	(0.016)	(0.010)
Plant-new SEIN pair fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.510	0.532	0.538	0.525
N	17,585	106,375	18,122	105,838

We include the same set of controls as in our wage (i.e., displaced women apply to female-led firms at higher

(a) Table 9: Female leadership in closing plant and wage changes.

Table 10
Quantity effects.

Ordinary least squares (OLS) regressions using an indicator for a female chief executive officer (CEO) as the dependent variable. *Female CEO* indicates that the top earner in the hiring firm (FIRMID) is female. The omitted race category is white. *Age* is worker age. *Wage* is the pre-closure annual wage. The pre-closure wage is the annual wage (defined in Table 2) two quarters prior to plant closure. *Tenure* is measured as the number of quarters that a worker has spent in the firm. All worker-level variables are measured two quarters prior to plant closure. *Manager* is defined as the highest paid employee in the plant. Standard errors are clustered by closing plant and are reported in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

Independent variable	OLS (1)	OLS (2)
Race = black	0.002	0.000
	(0.003)	(0.003)
Race = Asian	0.003	0.003
	(0.004)	(0.004)
Race = Hispanic	0.000	0.001
	(0.003)	(0.003)
Race = other minorities	0.002	0.002
	(0.003)	(0.003)
ln(*Age*)	0.014***	0.012***
	(0.003)	(0.003)
ln(*Wage*)	−0.008***	−0.008***
	(0.002)	(0.002)
Manager	−0.009	−0.013**
	(0.006)	(0.006)
ln(*Tenure*)	−0.002*	−0.002
	(0.001)	(0.001)
Female	0.034***	0.024***
	(0.003)	(0.002)
Plant fixed effects	Yes	Yes
New industry fixed effects	No	Yes
Adjusted R^2	0.004	0.034
N	160,642	160,642

(b) Table 10: Quantity effects.

Tables 9 and 10: Origin culture and worker sorting.

7.6 Table 11: Industry competitiveness and wage changes

Read:

- Table 11 splits workers by competitiveness of the closing plant's industry.
- The female CEO interaction is strongest in the most concentrated industries.
- The effect is much weaker or insignificant in more competitive industries.

Detailed interpretation: This table is central to the discrimination mechanism. In concentrated labor markets, firms have greater discretion to deviate from optimal wage-setting, so discriminatory wage gaps can persist. Female CEOs have the strongest impact where such discretion is greatest.

Economic message: The mechanism is not only productivity or amenities. The evidence is consistent with female leaders mitigating demand-side discrimination in wage setting.

Table 11
Industry competitiveness of closing plant and wage changes for displaced workers.

The dependent variable is the difference in the natural logarithms of the pre- and post-plant closure wage. The pre-closure wage is the annual wage (defined in Table 2) two quarters prior to plant closure and the post-closure wage is the annual wage four quarters following the plant closure. The omitted race category is white. *Age* is worker age. *Wage* is the pre-closure annual wage. *Tenure* is measured as the number of quarters that a worker has spent in the firm. All worker-level variables are measured two quarters prior to plant closure. *Manager* is defined as the highest paid employee in the plant. *Female CEO* indicates that the top earner in the hiring firm (FIRMID) is female. Industry competitiveness is measured using a Herfindahl index of employment at the two-digit standard industrial classification level. We split industry-years into four equal-size groups. Standard errors are clustered by state employer identification number (SEIN) and are reported in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

Independent variable	Competitive industry (1)	– – – – – – – – – – > (2)	(3)	Concentrated industry (4)
Race = black	−0.039***	−0.021*	−0.015**	−0.043***
	(0.005)	(0.012)	(0.006)	(0.011)
Race = Asian	0.004	−0.041**	−0.019*	0.030
	(0.009)	(0.019)	(0.010)	(0.024)
Race = Hispanic	−0.032***	−0.040***	−0.015	−0.021*
	(0.005)	(0.007)	(0.010)	(0.011)
Race = other minorities	−0.005	−0.019**	−0.013**	−0.025**
	(0.006)	(0.008)	(0.006)	(0.012)
ln(*Age*)	−0.077***	−0.059***	−0.068***	−0.080***
	(0.009)	(0.013)	(0.009)	(0.014)
ln(*Wage*)	−0.102***	−0.145***	−0.167***	−0.126***
	(0.006)	(0.011)	(0.032)	(0.017)
Manager	0.015	0.016	0.091**	0.019
	(0.015)	(0.017)	(0.043)	(0.030)
ln(*Tenure*)	−0.018***	−0.003	−0.007	−0.030***
	(0.003)	(0.005)	(0.004)	(0.007)
Female	−0.043***	−0.037***	−0.046***	−0.039***
	(0.006)	(0.005)	(0.007)	(0.008)
(*Female CEO*) × (*Female*)	0.018*	0.018	0.019	0.057***
	(0.011)	(0.014)	(0.019)	(0.022)
Plant-new SEIN pair fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.484	0.582	0.576	0.559
N	79,634	36,701	31,425	12,882

our data, we are also able to correct for endogenous

affect the wage gap between men and women, female

Table 11: Industry competitiveness of closing plant and wage changes for displaced workers.

8 Result map

- **Baseline wage-level result:** female leadership is associated with a smaller gender wage gap in the random worker sample.
- **Displacement penalty:** displaced women suffer 4–5 percent larger wage losses than comparable displaced men.
- **Main leadership result:** female leadership in the hiring firm mitigates the relative wage loss of women.
- **Critical mass:** majority female leadership among the top five earners is more powerful than one female leader.
- **Firm-level culture:** female CEOs matter more than female divisional leaders, implying a culture channel.
- **Origin culture:** women displaced from female-led plants already have better relative outcomes and gain less from female-led hiring firms.
- **Quantity effects:** displaced women are more likely to move to female-led firms, consistent with higher demand for female workers.
- **Discrimination mechanism:** effects are strongest in concentrated industries where wage-setting discretion is higher.

9 Short summary

- Tate and Yang study whether female leadership reduces gender wage gaps.
- They use LEHD and LBD administrative data to follow workers displaced by plant closures.
- The main sample compares workers from the same closing plant who move to the same hiring SEIN.

- Displaced women lose about 4–5 percent more wages than displaced men.
- The gap is significantly smaller when women hold leadership roles in the hiring firm.
- The effect is strongest when women are a majority of the top five leaders.
- Female CEOs matter more than female divisional CEOs, suggesting firm-wide culture.
- Effects are strongest for women from male-led plants and concentrated industries.
- The paper interprets female leadership as creating female-friendly cultures and reducing discrimination.

10 Conclusion

10.1 Core conclusion

Female leaders reduce the relative wage losses of displaced women. The evidence suggests that leadership diversity has benefits below the executive level by changing wage-setting practices and organizational culture.

10.2 Economic intuition

Plant closures force men and women to search for new jobs. When men and women from the same old plant move to the same new employer, they should face similar shocks and similar destination wage policies. The fact that women still lose more than men reveals a residual gender gap. Female leadership at the new employer reduces that residual gap, suggesting a difference in firm culture or wage-setting treatment.

10.3 Incremental contribution revisited

The paper contributes by combining a strong displacement design with administrative worker-firm data and leadership measures constructed from top earners. It moves the literature from documenting female underrepresentation to identifying a positive externality of female leadership for other women.

10.4 Policy implications

The results imply that increasing female representation in leadership may benefit women beyond those promoted. Diversity at the top can affect lower-level workers through culture, wage-setting, recruiting, and promotion norms.

10.5 Limitations

The leadership measures are based on pay rank, not job titles. The design is strong but not randomized; female leaders may still be associated with unobserved cultural changes. The data cover 23 LEHD states and a specific historical window. The analysis focuses on wage changes after plant closures rather than all labor-market transitions.

10.6 Final takeaway

Female leadership is not only symbolic. It is associated with materially better wage outcomes for other women, especially when workers are forced to make job changes and when labor markets allow employers discretion in wage setting.